Literature-Implied Answer: The Atomic Masa Subcase

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Status: literature-implied answer (partial subcase)

Source question

In Section 4, PDF page 9, of Wassermann's *Tensor Products of Maximal Abelian Subalgebras of C^*-Algebras* [1], the author asks whether $(C \otimes 1)^c = C \otimes B(H)$ in $B(H) \otimes_{\max} B(H)$ for every masa $C \subset B(H)$, and whether $C_1 \otimes C_2$ is maximal abelian there for every pair of masas. The paper calls the case

$$C_1 \cong C_2 \cong \ell^\infty(\mathbb{N})$$

particularly intriguing and observes that a positive solution of the Kadison–Singer problem would imply a positive answer for this product-masa case by its Corollary 3.

Identification

Marcus, Spielman, and Srivastava prove Weaver's KS_2 conjecture and an Anderson paving formulation, both known to imply a positive solution of the Kadison–Singer problem [2]; see their abstract and Question 1.1 on PDF pages 1–2. Therefore the standard diagonal masa $D = \ell^\infty(\mathbb{N}) \subset B(\ell^2(\mathbb{N}))$ has the extension property.

Wassermann's Theorem 2 now applies with $C = D$, $A_1 = A_2 = B(H)$, and the maximal norm, giving

$$(D \otimes 1)^c = D \otimes B(H) \quad \text{in } B(H) \otimes_{\max} B(H).$$

His Corollary 3, applied to $C_1 = C_2 = D$, gives

$$D \otimes D \text{ is a masa in } B(H) \otimes_{\max} B(H).$$

Thus both atomic diagonal subcases have a positive answer. This is a direct post-2013 literature implication, not a new mathematical solution, and the supporting paper does not identify Wassermann's question explicitly.

Scope and search evidence

The conclusion does not cover the non-atomic masa $L^\infty(0,1) \subset B(L^2(0,1))$, nor arbitrary pairs of masas. Searches of the run indexes for arXiv:0711.0131, masa, maximal tensor product, and the exact relative-commutant formulas found no earlier packet. A bounded external search found the source and Kadison–Singer literature but no primary source settling the remaining diffuse/arbitrary-masa questions.

References

[1] S. Wassermann, *Tensor Products of Maximal Abelian Subalgebras of C*-Algebras*, Glasgow Math. J. 50 (2008), 209–216; arXiv:0711.0131.

[2] A. W. Marcus, D. A. Spielman, and N. Srivastava, *Interlacing Families II: Mixed Characteristic Polynomials and the Kadison–Singer Problem*, Ann. of Math. 182 (2015), 327–350; arXiv:1306.3969.